Inputs via Switches

Digital Inputs

All switches generate a digital signal i.e. ON or OFF, HIGH or LOW, 1 or 0

As a result we connect them to the digital IO pin to read their state.

Wiring

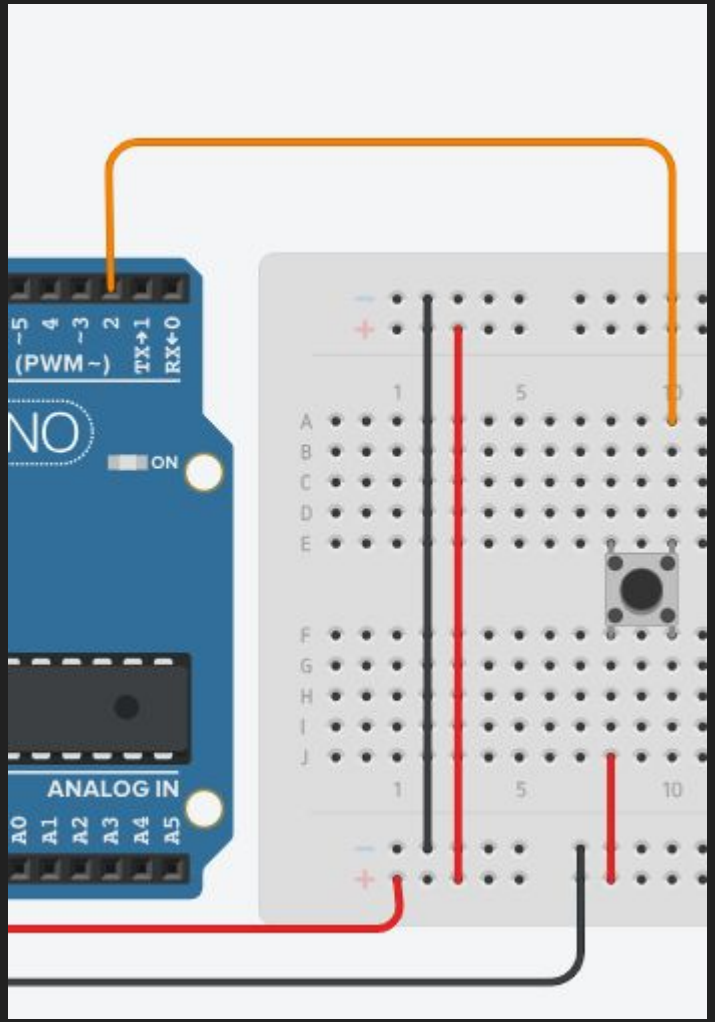
There’s a problem with wiring a button this way.

When the button is pressed, 5V is connected to

pin 2 and when read will generate a HIGH(1) value.

However, when not pressed, pin 2 is not connected to anything and so pin 2 will read

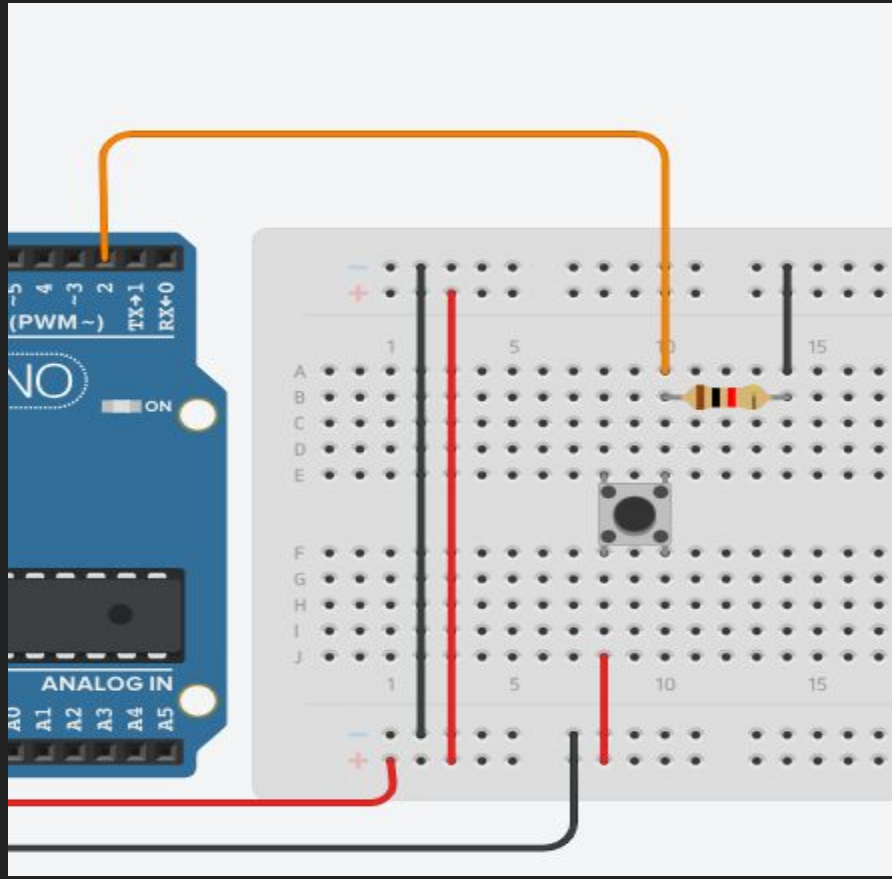
floating or changing/unpredictable values.



Solution-Pull Down or Pull Up Resistors

A pull up or pull down resistor must be connected via 1K-10K resistors. In this example we are using a pull down resistor valued at 1KOhms.

When not pressed, pin 2 is connected to ground and will read a LOW(0). When the button is pressed power will pass through the button and flow towards the pin 2 and avoid GND and a short circuit because of the high valued resistor. Electrons always want to follow the path of least resistance. As a result when the button is pressed pin 2 reads HIGH(1).

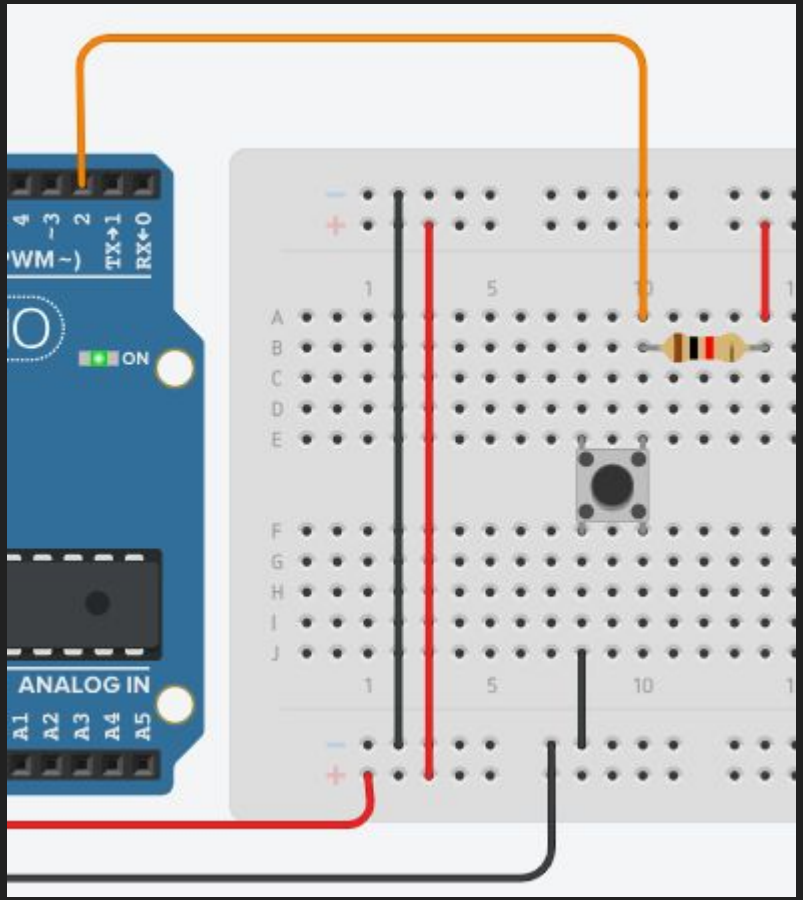


Pull Up Resistor

The last slide showed an example of a pull down resistor. Here we see a pull up resistor.

When not pressed pin 2's value is pulled up to a HIGH(1) and when pressed it is pulled LOW(0).

The resistor is required to avoid a short circuit/over supply of current to pin 2.



Code

```
int val=0;
void setup()
{
  Serial.begin(9600);
  pinMode(2, INPUT);

}

void loop()
{
  val=digitalRead(2);
  Serial.println(val);
  delay(100);
}
```

Assignment(s) /30TI

Create a circuit with three push buttons and 3 LEDs. When the first button is pressed the first LED turns on. When the first button is not pressed the same LED turns off.

When the second button is pressed the second LED flashes with a quarter second delay. Not pressing the button means the flashing stops.

When pressing the third push button all the LEDs turn on one after the other with a quarter second delay in between and then turn off one after the other with a quarter second delay. This continues indefinitely until the third button is pressed again.

https://www.tinkercad.com/things/0uXArmTVCye-exquisite-inari-duup